Forward-Gap Layers of Consecutive Primes: Window-Free Diagnostics of the Gallagher Limit and a Derived Spatial Crossover at $g \approx \log X$

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Abstract

Under the Hardy–Littlewood conjectures, gaps between consecutive primes near X, normalized by $\log X$, become exponential with rate 1 (the Gallagher limit), yet fitted rates near 1.2 persist. With a verified census of prime gaps at 20 scales up to 10^{10}, we show the fitted rate depends more on the fitting window than on X: across 36 specifications its spread at one scale exceeds its change over four decades, and a published value is recovered by specification choice alone. On nineteen disjoint intervals from differencing the census, window-free moment ratios and a Kolmogorov discrepancy approach their exponential values at 17 of 18 steps, with deviations decaying as a fractional power of $\log X$. A parameter-free density model predicts a crossover at $g \approx \log X$ in the mean position of primes carrying gap g; the census confirms its location to 4%, with an amplitude 1.28 times the prediction.

Keywords: consecutive prime gaps; Gallagher limit; Hardy–Littlewood conjecture; Poisson model; moment ratios; finite-scale corrections; scaling collapse
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1 Introduction

1.1 Motivation

Let $p_1 < p_2 < \cdots$ be the primes and $d_n = p_{n+1} - p_n$. The Prime Number Theorem fixes the average gap near X at $\log X$, and the Hardy–Littlewood prime-pair conjecture [12] predicts the frequency of each even gap up to the singular-series factor $H(g)$ that encodes the small prime divisors of g. Gallagher [9, 10] showed that a uniform form of the Hardy–Littlewood conjectures forces the primes in short intervals to follow a Poisson law, so that gaps normalized by $\log X$ are asymptotically exponential with rate exactly 1. We refer to this as the Gallagher limit.

The computational record has never quite agreed with it. Brent [2] fitted the decay of gap frequencies and found a rescaled rate persistently above unity. The frequency of a given gap up to the singular series is the object of the jumping-champion analysis of Odlyzko, Rubinstein and Wolf [20], and Wolf [24, 25] turned it into the quantitative conjecture we test here, together with the finite-X heuristic $C_W(X) = \pi(X)\log X/X$ that tends to 1 slowly from above. Fitting that conjecture, Wolf reports a rescaled rate falling from 1.187 to 1.136 as X runs from 2^{28} to 2^{48} [24]. Gap computations now reach 4×10^{18} [21], yet whether a rate near 1.2 is a constant or a decaying artifact has not been settled, largely because the question has been posed in terms of a fitted slope whose definition is never fixed: neither the

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range of g nor the weighting is usually stated, and Section 4.4 shows that those choices move the answer further than four decades of computation do.

Two recent developments sharpen the question. Cohen [3] proved that the conjecture of Oliveira e Silva et al. on power sums of gaps holds if and only if the kth moment of the first n gaps is asymptotic to the kth moment of an exponential law with mean $\log n$, while the distribution itself is not exponential; this separates moment convergence from distributional convergence and suggests testing the former directly. Lemke Oliver and Soundararajan [14] showed that finite-range regularities in consecutive primes can be large, structured and slow to disappear, which is a warning against reading any finite-range fit as an asymptotic statement.

1.2 Objective and research questions

The objective of this paper is to say something about the approach to the Gallagher limit that does not depend on analyst choices, using a single verified computation. We ask three questions.

Q1 Is the fitted rescaled decay rate $C_{\mathrm{fit}}(X)$ well enough defined that extending it to larger X can settle the plateau question? (Section 4)

Q2 Do window-free local statistics of the gap distribution move toward the exponential law as X grows, and at what rate? (Section 5)

Q3 Does the spatial position of the primes carrying a given gap obey a scaling law that can be derived, without free parameters, from the prime density and the Hardy–Littlewood envelope? (Section 6)

1.3 Contributions

1. A complete census of consecutive prime gaps at 20 scales to 10^{10}, organized by the forward gap of each prime, with released code, reproduced exactly by an independent implementation on all 2,166 layer counts to $X = 9{,}999{,}999{,}967$ (Section 3).

2. A demonstration that $C_{\mathrm{fit}}(X)$ depends on the fitting specification more than on X, so that the published plateau value is accounted for by window choice alone and no extension of the range can decide the question in these terms (answer to Q1).

3. Three window-free local diagnostics with bootstrap uncertainty, obtained by differencing the census, which move toward the Gallagher prediction at 17 of 18 steps and by 10 to 21 standard errors end to end, with deviations decaying as a fractional power of $\log X$ (answer to Q2).

4. A parameter-free derivation of the spatial scaling form $R = 1 + \Phi(g/\log X)/\log X$ and its crossover at $g \approx \log X$, confirmed to 4% in location across four decades, together with a quantified failure of the same model to reproduce the amplitude (answer to Q3).

We claim no novelty for the partition itself, which is an elementary labeling device.

2 The census

2.1 Forward-gap layers

For $N \geq 2$ and even g define the forward-gap layer

$$L(g;N) = \{p_n : 2 \leq n \leq N,\ d_n = g\}. \tag{1}$$

Each p_n with $2 \leq n \leq N$ has exactly one outgoing gap, so the nonempty layers are pairwise disjoint with union $\{p_2, \ldots, p_N\}$ (Lemma 1). The empirical frequency of gap g at scale $X = p_N$ is $f(g;N) =$

$|L(g;N)|/(N-1)$. Classifying by the outgoing rather than the incoming gap means a prime's layer never has to be revisited when the computation is extended, which is what makes one cumulative census usable at 20 nested scales. It also attaches each gap to a definite prime, which is what the spatial statistic of Section 6 needs.

2.2 Boundary convention

The layer of p_n is determined by p_{n+1}, so a sieve that stops exactly at X cannot classify its last prime and, less obviously, cannot report layer occupancies at $N = \pi(X)$ at all; a checkpoint placed exactly at $\pi(X)$ is then silently skipped. The remedy is to sieve past X and discard the overshoot once the final gap has been recorded. Bertrand's postulate gives a bound that needs no tables: for every prime $p \le X$ the next prime is at most $2p \le 2X$, so sieving to $2X$ always suffices (Lemma 2). In practice a much smaller margin is enough: the maximal gap below 10^{10} is 354 [19, 21], and the production run sieves to $10^{10} + 10^3$. We record the point because the error is invisible in the prime count and any reimplementation will meet it.

2.3 Arithmetic factor

Following Hardy and Littlewood [12], the frequency of gap g is modulated by

$$H(g) = \prod_{q \mid g,\ q>2} \frac{q-1}{q-2}, \qquad C_2 = 2\prod_{q>2}\Big(1 - \frac{1}{(q-1)^2}\Big) \approx 1.3203236, \tag{2}$$

the products over odd primes q [5]. The adjusted frequency $f(g;N)/H(g)$ is expected to follow a smooth exponential envelope in g; the envelope is an asymptotic statement about a smoothed distribution, not a pointwise one, and Section 4 is largely a study of what goes wrong when it is treated as the latter. The period-six oscillation that $H(g)$ imposes on raw gap counts is discussed by Ares and Castro [1] and Wolf [25].

3 Methods, verification and reproducibility

3.1 Computing the census

Primes were generated by a bit-packed segmented sieve of Eratosthenes over the odd integers (`fgp_sieve.c`), of the standard form described by Crandall and Pomerance [5]. The segment window is 256 KiB, chosen to stay in L2 cache; base primes below 10^5 are sieved once. Memory is $O(\sqrt{X})$ plus one window, a few megabytes independent of X. Primes are streamed in increasing order, each accumulated into the layer indexed by its outgoing gap, and layer occupancies, layer prime-sums and layer sums of squares are snapshotted at 20 checkpoints. The layer prime-sums are exact: their total to 10^{10} is about 2.2×10^{18}, below the unsigned 64-bit bound by a factor of eight, so the spatial means are exact rationals; the sums of squares are accumulated in 128 bits. The checkpoint schedule is 20 geometrically spaced values of N from 200,000 to $\pi(10^{10})$, with $N = 10^6$, 9×10^6 and 10^8 snapped onto the grid so that scales in the earlier literature are directly comparable. The run to $10^{10} + 10^3$ takes about 14 seconds on one core of a desktop machine.

3.2 Verification

Prime counts and indices. $\pi(10^7) = 664{,}579$ and $\pi(10^{10}) = 455{,}052{,}511$, matching tabulated values [5, 7]; $p_{200000} = 2{,}750{,}159$, $p_{10^6} = 15{,}485{,}863$, $p_{10^8} = 2{,}038{,}074{,}743$; the largest prime below 10^{10} is 9,999,999,967.

Maximal gap. The largest gap below 10^{10} is 354, following 4,302,407,359, reproducing the first-occurrence tables of Nicely [19] and consistent with [21].

Local Prime Number Theorem. On each differenced interval the mean gap agrees with $\log X_{\text{eff}}$ to within 5×10^{-4} relative (Section 5).

Independent reimplementation. A second census, written from scratch in NumPy with one byte per integer, segments of 10^8, and gaps taken by differencing arrays rather than by a running pointer (`verify.py`), recomputes every layer count at every checkpoint over the whole range, to $X = 9{,}999{,}999{,}967$. All 2,166 counts agree exactly (Table 1); the run takes 121 seconds. The two programs share no code, no data layout and no language, so this removes the single-implementation caveat at every one of the 20 scales. The verifier, like the census, sieves past its limit by a margin, since the layer of p_N at $X = 10^{10}$ is fixed by $p_{N+1} = 10{,}000{,}000{,}019$, and it aborts rather than truncating if a gap exceeds the width of its histogram.

[Table 1 about here.]

Self-test suite. The released `selftest.py` runs forty-five checks with no network access, confronting the pipeline with facts established independently of it: tabulated $\pi(10^k)$ and p_n; the twin-prime counts $\pi_2(10^k)$, which the $g = 2$ layer must reproduce exactly (at 10^{10}, 27,412,679); the maximal-gap record; the telescoping identity, which recovers $p_{N+1} = 10{,}000{,}000{,}019$ from the layer sums; the arithmetic factor $H(g)$ against symbolic factorization; the Poisson fit against a general-purpose optimizer; the bootstrap standard error against the delta method; and the numerical density model against the analytic expansion of Appendix A. Checks of this kind are not a formality. During development the suite detected an implementation of $H(g)$ that did not remove the factor 2 before trial division and so assigned a wrong factor to most even gaps below 200; because $\log H(g)$ enters the Poisson fit as an offset, it shifted every C_{fit} by 0.011 to 0.024, which is less than the width of the specification envelope of Section 4.4 and left every plot looking reasonable. The local diagnostics do not use H, and it cancels in R. The suite also exposed an accumulator that would silently discard gaps wider than its histogram (harmless below 10^{10}, where the maximal gap is 354, but not beyond roughly 7×10^{11}); both programs now abort instead. Appendix C lists every assumption the results rest on, marked as verified, stated, or limiting.

3.3 Software and data

The released repository contains the sieve (`fgp_sieve.c`), a pure-Python census generator producing byte-identical output for readers without a C toolchain (`make_census.py`, a few minutes to 10^{10}), a Python package `fgp` (modules `core`, `frequency`, `local`, `spatial`, `verify`, `figures`, `tables`), a single entry point `run_all.py` that regenerates every table, figure and number quoted in this paper from the census, and a Lean 4 formalization of the elementary statements of Appendix B. The primary dataset `fgp_layers.csv` has 2,166 rows with columns checkpoint, N, X, g, count, sum and sum of squares. Every table in this paper is written by the code as a LaTeX fragment and every number in the text by the same code, so the manuscript cannot drift from the computation. The pipeline is

```
cc -O3 -funroll-loops -o code/fgp_sieve code/fgp_sieve.c -lm
./code/fgp_sieve 10000001000 455052511 > data/fgp_layers.csv
python3 code/selftest.py data/fgp_layers.csv
python3 code/run_all.py data/fgp_layers.csv --out paper \
       --verify-limit 10000000000
```

Code and data are released into the public domain (CC0).

4 The fitted decay rate cannot settle the question

4.1 The Wolf benchmark is a moment estimator

Wolf's heuristic [24, 25] predicts, for the number $\tau(g;X)$ of gaps g below X,

$$\tau(g;X) \approx C_2 H(g)\,\frac{\pi(X)^2}{X}\exp\!\Big(-g\,\frac{\pi(X)}{X}\Big). \tag{3}$$

The rescaled decay rate $C(X)$ is the slope in g of $\log\bigl(f(g;X)/H(g)\bigr)$ multiplied by $\log X$, and the Gallagher limit is $C(X)\to 1$. We write $C_{\mathrm{fit}}(X)$ for its regression estimate and $C_W(X)=\pi(X)\log X/X$ for the benchmark in (3).

Wolf fits (3) in the variable $u=g\,\pi(X)/X$ rather than in g, so his slope $s(X)$ and our λ are related by $\lambda = s\,\pi(X)/X$, that is

$$s(X)=\frac{C_{\mathrm{fit}}(X)}{C_W(X)}. \tag{4}$$

Both s and C_{fit} tend to 1 under the Gallagher limit, and they differ at finite X by exactly the factor C_W. We record (4) because it is what makes the values published in that normalization directly comparable with this census, and we use it in Section 4.4.

One identity is worth recording. The mean gap below X is $(p_{N+1}-p_2)/(N-1)\approx X/\pi(X)$ (Lemma 3). If the normalized gaps $g/\log X$ followed an exponential law with rate C, the method-of-moments estimator of C would be the reciprocal of the mean normalized gap, that is $\pi(X)\log X/X = C_W(X)$. So C_W is not an external prediction: it is the moment estimator of the same parameter that C_{fit} estimates from the tail slope, and the excess $\Delta(X)=C_{\mathrm{fit}}-C_W$ compares two estimators of one quantity. It measures departure from exponentiality, which is what Cohen's criterion [3] isolates through moments, and it points to where a window-free diagnostic should be sought.

4.2 Fitting procedure

We fit the Poisson log-linear model

$$\mathbb{E}\,|L(g;N)| = \exp\bigl(a+\log H(g)-\lambda g\bigr) \tag{5}$$

by iteratively reweighted least squares [16], with $\log H(g)$ as a fixed offset. This is the appropriate likelihood for counts, handles the sparse tail without the bias that logarithms of small counts introduce, and gives an asymptotic standard error from the Fisher information. The primary specification, fixed before the analysis, fits over even g with $4\le g\le 3\log X$; then $C_{\mathrm{fit}}=\lambda\log X$. We exclude $g=2$ because it is the one layer for which the singular-series correction is trivial and the departure from the smooth envelope largest at accessible X.

4.3 Results

Table 2 gives the 20-scale census under the primary specification. $C_{\mathrm{fit}}(X)$ falls from 1.2483 at $X=$ 2,750,159 to 1.1460 at $X=$ 9,999,999,967 and decreases at every one of the nineteen consecutive steps, by at least 1.8 combined standard errors (a figure that treats the nested fits as independent and therefore overstates the significance); $C_W(X)$ falls more slowly, from 1.0783 to 1.0478; and $\Delta(X)$ falls from 0.1700 to 0.0982. Taken alone, this reads as the value near 1.2 being a finite-X effect (Figure 1).

[Table 2 about here.]

[Figure 1 about here.]

4.4 The specification grid

We recomputed $C_{\mathrm{fit}}(X)$ at all scales under 36 specifications: three estimators (the Poisson model; weighted least squares on $\log(f/H)$ with weights $|L(g;N)|$; unweighted least squares), two lower cutoffs ($g\ge 2$, $g\ge 4$) and six upper cutoffs ($g\le c\log X$, $c=1,\dots,6$). Each is defensible and versions of each appear in the computational literature [2, 24, 25]. The window runs to $c=6$ because Wolf's published fit, described as taken over the linear portions of a semi-logarithmic plot, turns out to correspond to c near 6; a grid stopping at $c=4$ would not cover the literature it is meant to bracket. Table 3 and Figure 2 summarize the grid.

[Table 3 about here.]

[Figure 2 about here.]

Four things follow. First, the spread of C_{fit} across specifications at a single X lies between 0.206 and 0.352, whereas the total change under the primary specification over four decades is 0.102: the analyst's choice moves the statistic further than the computation does. Second, monotonicity is a property of the specification, not of the data: all 36 decrease on net, but only 20 decrease at every one of the 20 scales, so whether one reports a clean monotone approach depends on the window. Third, the value at $X = 10^{10}$ ranges from 0.9999 under the narrowest window to 1.2150 under the widest, spanning the Gallagher value itself: the adjusted frequency is flatter than exponential at small g and steeper in the tail, so the "decay rate" is largely a choice of which part of a non-exponential curve to call the exponent. Fourth, and most directly, the published value is recovered from our own data by a change of specification. Wolf [24] reports s falling from 1.187 to 1.136 as X runs from 2^{28} to 2^{48}. At our scale nearest the start of that range, $X = 325{,}619{,}971$, the primary specification gives $s = 1.1114$ while the grid spans $[0.9382, 1.1910]$, so the published value falls inside it; the closest specification, $g \geq 2$ with $g \leq 6 \log X$ fitted by unweighted least squares, gives $s = 1.1869$, which reproduces the published 1.187 to all three decimals reported. That combination is what Wolf's own description of the procedure implies. The published number is therefore accounted for by the specification alone: it need not indicate a constant, and our lower values under a different specification need not refute one.

We conclude that the question "does $C_{\mathrm{fit}}(X)$ converge to 1?" is not well posed, and that no extension of the computational range will make it so. This answers Q1.

5 Window-free local diagnostics

5.1 Differencing the census

Two problems afflict Section 4: the window, and the fact that every cumulative statistic over $[2, X]$ is a mixture over scales, since $\log t$ varies by a factor above twenty across that interval. Because the census stores cumulative layer occupancies at 20 checkpoints, the difference of consecutive snapshots is the exact gap distribution on $I_k = (X_{k-1}, X_k]$. Consecutive checkpoints are a factor of about 1.5 apart, so $\log t$ varies by under two percent within I_k. The nineteen differenced distributions are local, and they are supported on disjoint sets of primes. The primes are deterministic, so "independence" here means independence under the multinomial resampling model of Section 5.3; that is the sense in which the significance statements below are made.

On each I_k we compute, with no window and no estimator, the mean gap m_1, the moment ratios

$$E_2 = \frac{m_2}{2m_1^2}, \qquad E_3 = \frac{m_3}{6m_1^3}, \tag{6}$$

which equal 1 for an exponential law (Lemma 4), and the Kolmogorov discrepancy

$$D = \sup_g \left| S(g) - e^{-g/m_1} \right| \tag{7}$$

between the empirical survival function and the exponential with the same mean. D is a discrepancy, not a test statistic: the comparison law is fitted from the same data, so the Kolmogorov null distribution does not apply, and we attach a bootstrap standard error instead. Gaps take values in $2\mathbb{Z}$, so e^{-g/m_1} evaluates the continuous law at the left endpoint of the cell $(g, g+2]$; we also report the midpoint convention $e^{-(g+1)/m_1}$. Because $S(g)$ sums over gaps, the oscillation $H(g)$ is averaged out and no singular-series correction is needed here.

The effective scale of I_k is $\log X_{\mathrm{eff}} = \big(X_k(\log X_k - 1) - X_{k-1}(\log X_{k-1} - 1)\big)/(X_k - X_{k-1})$, the mean of $\log t$ over the interval. (The density-weighted mean $(X_k - X_{k-1})/(\mathrm{li}\, X_k - \mathrm{li}\, X_{k-1})$ differs from it by under 0.002 on every interval.)

5.2 Results

Table 4 gives the local diagnostics and Figure 3 plots them. The first observation is a validation: the local mean gap agrees with $\log X_{\text{eff}}$ to within 5×10^{-4} in relative terms on every interval (ratio in $[0.99959, 1.00020]$). This is the Prime Number Theorem reproduced locally, and a sharper check on the census than matching tabulated $\pi(X)$ values [7], because it tests the gap bookkeeping at every scale.

[Table 4 about here.]

[Figure 3 about here.]

The substantive result is that E_2 rises from 0.84606 to 0.88815, E_3 from 0.68323 to 0.76691, and D falls from 0.05643 to 0.04505. Prime gaps at these scales are under-dispersed relative to the exponential law ($E_2 < 1$: more regular than a Poisson process), and the under-dispersion is shrinking. This is the finite-X counterpart of Cohen's asymptotic statement [3]: the moments approach their exponential values while the distribution remains non-exponential throughout the range.

Under the midpoint convention D falls from 0.10569 to 0.08102, roughly double in absolute terms with the same monotone decrease; the absolute level of D is convention-dependent and only its trend should be read.

On the rate of approach, the deviations are not well described by $c/\log X$. A log-log regression of each deviation on $\log X_{\text{eff}}$ gives exponents 0.810 ± 0.011 for $1 - E_2$, 0.782 ± 0.012 for $1 - E_3$ and 0.592 ± 0.017 for D: fractional powers of $\log X$. The exponent for D depends on the continuity convention (the midpoint form gives 0.67 ± 0.01), which is one more reason to prefer the moment ratios. Extrapolated, $1 - E_2$ would still be about 0.059 at $\log X = 50$ and 0.019 at $\log X = 200$. We are extrapolating from under one decade in $\log X$ and do not offer these as predictions; they quantify how far out of reach a decisive computation is.

5.3 Uncertainty and significance

We attach a nonparametric bootstrap standard error to each statistic by resampling the layer counts on each interval from the multinomial distribution they define (2000 resamples per interval) [8]. Because the intervals are disjoint, the standard error of a consecutive difference is the root sum of squares of the two. Table 5 gives six representative intervals and Table 6 the step tests.

[Table 5 about here.]

[Table 6 about here.]

Three statements can be made. Of the 18 consecutive steps, 17 move each of E_2, E_3 and D toward the Gallagher prediction. The single exception is the first step, from I_1 to I_2, where E_2 falls ($z = -0.12$), E_3 falls ($z = -0.20$) and D rises ($z = -0.88$): all within one standard error of no change, at the smallest and noisiest scales in the census. Where the steps do move in the predicted direction they mostly do so significantly: the median z is 4.5 for E_2, 3.7 for E_3 and 2.8 for D. The end-to-end change is large relative to its own uncertainty: $z \approx 21$ for E_2, 17 for E_3 and 10 for D. A Cramér–von Mises distance W^2, which weights the whole distribution rather than its point of maximal deviation, decreases at all 18 steps.

What we claim is therefore this. Three statistics that involve no analyst choice move toward the Gallagher prediction across four decades, by 10 to 21 standard errors end to end, with a single within-noise exception at the smallest scale. That is evidence for convergence of a kind the fitted rate cannot supply. It is not proof, and it does not exclude a limit differing slightly from the exponential one. This answers Q2.

6 Spatial localization: a derived crossover

6.1 The relative spatial mean

Define the layer mean $\mu(g;N) = |L(g;N)|^{-1} \sum_{p \in L(g;N)} p$, the overall mean $\mu(N) = (N-1)^{-1}(p_2 + \cdots + p_N)$, and

$$R(g;N) = \frac{\mu(g;N)}{\mu(N)}. \tag{8}$$

R measures whether the primes carrying gap g sit on average earlier ($R < 1$) or later ($R > 1$) in the sequence than a typical prime up to the Nth. It is a direct empirical mean, with no fitting, no window and no arithmetic correction, and since the layer prime-sums are exact integers it is reproducible to full precision. The crossover $g^*(X)$ is the gap at which $R = 1$.

The census also stores the layer sums of squares, which is what makes a standard error possible. Since $\mu(N)$ contains $\mu(g;N)$, the two are correlated; writing $w = |L(g;N)|/(N-1)$ and splitting the primes into layer g (count n_g, mean μ_g, variance σ_g^2) and the rest (count n_r, mean μ_r, variance σ_r^2), the delta method gives

$$\mathrm{Var}(R) = (1-w)^2 \Big[\frac{\mu_r^2 \sigma_g^2}{n_g} + \frac{\mu_g^2 \sigma_r^2}{n_r} \Big] \Big/ \mu(N)^4, \tag{9}$$

every ingredient of which is an exact integer moment. As in Section 5, these are standard errors of a sampling model in which layer membership is random: the primes are deterministic, so (9) quantifies the arithmetic fluctuation of a layer rather than a probability about the primes. The released self-test checks (9) against a direct bootstrap over the actual primes of a layer at a scale small enough to enumerate them.

6.2 A parameter-free density model

Take the density of primes near t to be $1/\log t$ and, following (3) with $\pi(t)/t$ replaced by $1/\log t$, the conditional probability that a prime near t carries outgoing gap g to be $(C_2 H(g)/\log t)\exp(-g/\log t)$. The layer-$g$ density is the product, and $C_2 H(g)$ cancels identically in a ratio of means. Writing $w_g(t) = \exp(-g/\log t)/(\log t)^2$,

$$R_{\mathrm{model}}(g;X) = \frac{\int_2^X t\, w_g(t)\, dt \,/\, \int_2^X w_g(t)\, dt}{\int_2^X t/\log t\, dt \,/\, \int_2^X dt/\log t}. \tag{10}$$

The model has nothing to tune, so agreement with the data is a test rather than a fit. Substituting $t = e^{L-v}$, $L = \log X$, expanding to first order in $1/L$ and writing $a = g/L^2$ gives (Appendix A)

$$R_{\mathrm{model}} = 1 + \frac{a}{2} - \frac{1}{2L} + O(L^{-2}). \tag{11}$$

Setting $R = 1$ gives $a = 1/L$, that is $g^* = \log X + o(\log X)$: the crossover sits at the mean gap. In $x = g/\log X$ the correction takes the scaling form

$$R(g;N) = 1 + \frac{\Phi(x)}{\log X} + o\Big(\frac{1}{\log X}\Big), \qquad \Phi(1) = 0, \tag{12}$$

with $\Phi(x) = (x-1)/2$ to first order. Both the abscissa and the amplitude must be rescaled, the latter by $\log X$.

The replacement of $\pi(t)/t$ by $1/\log t$ is a simplification. The two differ by the factor $C_W(t) \approx 1.05$ at these scales, which is of the same order as the $O(1)$ corrections discussed below, so we evaluate (10) numerically for both envelopes (the released `spatial.py` uses a quadrature stable at arbitrarily large L). Figure 4 compares both versions of the model with the census at three scales.

[Figure 4 about here.]

6.3 The crossover

Because $R(g)$ is nearly flat near the crossing and single layers carry arithmetic fluctuations, g^* is located by a linear regression of R on g and solved for $R = 1$. The regression window is an analyst choice. The primary locator uses the scale-invariant window $0.6 \le g/\log X \le 1.4$, which is the same window in scaling units at every scale and matches the linear form of Φ to first order. It is a weighted least-squares fit using the standard errors (9), which also yields a standard error for g^* by the delta method. For comparison we report unweighted fits over the windows $[0.7, 1.3]$, $[0.6, 1.4]$ and $[0.5, 1.5]$ and over the bands $|R-1| < b$ for $b \in \{0.02, 0.03, 0.05\}$. Table 7 and Figure 5 give the results.

[Table 7 about here.]

[Figure 5 about here.]

Two conclusions are robust to the locator. The crossover rises with X (from 14.13 to 22.37) and tracks $\log X$: $g^*/\log X$ lies between 0.913 and 0.986 over all scales and all six locators. The deficit $\log X - g^*$ is $O(1)$, lying between 0.48 and 1.06.

The error bars change what can be claimed, in both directions. At the largest scale, $g^* = 22.37 \pm 0.27$ against $\log X = 23.026$, so $g^*/\log X = 0.9715 \pm 0.0116$ and the naive reading $g^* = \log X$ is excluded at 2.5 standard errors: the crossover sits slightly but detectably below the mean gap. In the other direction, no trend in the deficit can be claimed. The band locators show why: $b = 0.03$ produces an upward drift, $b = 0.02$ a downward one, and the scale-invariant windows none. More fundamentally, the 20 scales are nested, since the census at X_k contains every prime used at X_{k-1}, so the g^* series is strongly autocorrelated and a trend test that treats the scales as independent is invalid. Unlike the frequency statistics of Section 5, the spatial statistic cannot be localized to fix this: R acquires its dynamic range from the variation of $\log t$ across the whole interval $[2, X]$, and on one differenced interval R varies by under 0.005 against 0.38 cumulatively. Nesting is intrinsic to the statistic, and we make only single-scale statements about it.

Evaluating the model (10) numerically gives a deficit of 1.7149 at $\log X = 23$, decreasing to 1.5167 at $\log X = 230$ and to 1.5004 at $\log X = 10^4$, so $g^* = \log X - 3/2 + o(1)$ for the model. The census gives 0.66 ± 0.27 at the largest scale, which excludes $3/2$ at 3.2 standard errors. The model therefore predicts the leading term correctly and the $O(1)$ term incorrectly, and this is a measurement rather than an impression; the $\pi(t)/t$ envelope does worse still (deficit 2.78 at $\log X = 23$).

6.4 The scaling collapse

Figure 6 tests (12). Panel (a) shows R against g: the curves are distinct and the crossing moves rightward with X. Panel (b) rescales the abscissa alone: the crossings collapse onto $x \approx 0.96$, but the profiles do not, since the amplitude of $R-1$ shrinks with X. Panel (c) plots $\log X\,(R-1)$ against x, the full rescaling the model requires: the 20 curves collapse onto a single increasing profile with $\Phi(1) \approx 0$. The residual spread in panel (c) is a downward drift of the slope from about 0.91 to 0.78 across four decades (fitted over $0.3 \le x \le 1.5$), consistent with the $O(1/\log^2 X)$ terms neglected in the derivation. We regard the collapse as established for $x \lesssim 2$ and suggestive beyond it.

[Figure 6 about here.]

Panel (c) also exposes the model's limitation. In the collapsed variable the model profile at $\log X = 23$ has slope 0.62 (tending to the first-order value $1/2$ as $L \to \infty$) whereas the census profile has slope 0.78 at the same scale. Fitting both over the same window $0.3 \le x \le 1.5$ at every scale, the ratio of census slope to model slope is 1.28 ± 0.04 (mean and standard deviation over the 20 scales, range $[1.20, 1.39]$), with a total drift across the range smaller than that standard deviation. For the $\pi(t)/t$ envelope the ratio is 1.12 ± 0.04. Quoting residuals in R, as Figure 4(b) does, makes the agreement look better than it is, because $R-1$ is itself $O(1/\log X)$. The correct statement is that the model predicts the location

of the crossover to about 4% and the form of the scaling law, and under-predicts the amplitude by a scale-independent factor. A stable multiplicative discrepancy is what one expects if the mechanism is right and the envelope shape is wrong, and Section 5 has already shown that the gap distribution at these scales is not the exponential envelope assumed in (10). Whether feeding the measured under-dispersion back into (10) recovers the amplitude is an open question we have not pursued. This answers Q3.

7 Discussion

The two diagnostics share a mechanism: as X grows the typical gap $\log X$ grows with it, and both the frequency envelope and the spatial distribution of a fixed g are reshaped relative to that moving scale. What differs is how cleanly each can be measured.

On the frequency side the contribution is a caution and a replacement. The caution is that the fitted decay rate around which the plateau question has been posed [2, 20, 25] is specification-dependent to a degree exceeding the effect under study. The replacement is the local diagnostics of Section 5. Our reading is that the value near 1.2 is best understood neither as a constant nor as a slowly decaying estimate of unity, but as one summary of a distribution that is not exponential at accessible X, measurably under-dispersed ($E_2 \approx 0.88815$ at 10^{10}), whose departure from exponentiality decays as a fractional power of $\log X$. This is consistent with Cohen [3], with the refinements of the Cramér model [4] by Granville [11] and by Montgomery and Soundararajan [17, 18], which predict slow and structured approach for several classical gap statistics, and with the warning of Lemke Oliver and Soundararajan [14]. None of it bears on the bounded-gap theorems of Zhang [26] and Maynard [15], which concern the extreme lower tail rather than the bulk studied here; for the wider background see Soundararajan [22].

On the spatial side the crossover appears to be a new empirical observation and comes with a derivation. That $C_2 H(g)$ cancels tells us $R(g;N)$ is insensitive to the arithmetic of g and reflects only the interaction between the growth of $\log t$ and the envelope. This bounds the phenomenon's depth: the crossover is a consequence of the envelope and the prime density, not an independent arithmetic fact. Its interest is diagnostic. R is measured without fitting, reproduces exactly across implementations, and its crossover location converges to its scaling form visibly faster than the frequency statistics converge to theirs. The amplitude discrepancy of Section 6.3 is, we think, the most useful thing the spatial statistic has produced so far: it is a quantitative, scale-stable measurement of how far the actual gap law is from the exponential envelope, obtained without any window.

8 Limitations

The Gallagher limit is not established. Section 5 reports approach with end-to-end significance of 10 to 21 standard errors, but at fractional-power rates: the extrapolated deviation at $\log X = 50$ is still about 0.059 in E_2. Nothing here distinguishes the Gallagher limit from a limit differing from it by a few percent.

One step is not monotone. The transition from I_1 to I_2 moves all three statistics marginally the wrong way, within one bootstrap standard error. We report it rather than trimming the smallest interval.

The fractional exponents are descriptive. They are least-squares fits over $\log X$ from 15 to 23 and we have no derivation for them. The exponent for D depends on the continuity convention.

The crossover scales are nested, so only single-scale statements are made. The deficit is measured to about ± 0.27 at the largest scale, which is enough to exclude both 0 and the model's $3/2$ but not to establish its value or any trend. The spatial statistic cannot be localized into independent samples the way the frequency statistics can (Section 6.3). The model's $3/2$ rests on numerics of a heuristic, not on an analytic second-order evaluation.

The density model under-predicts the amplitude of $R-1$ by a factor of 1.28 ± 0.04, with no trend across scales. It assumes the exponential envelope exactly and uses $1/\log t$ for the prime density; it

should also fail in the far tail $g \gtrsim 1.5 \log X$, where layers are sparse and the maximal-gap heuristics of Kourbatov and Wolf [13] take over, and Figure 4 shows that it does.

Independence is model-based. The primes are deterministic. Independence of the differenced intervals holds under the multinomial resampling model, and the z-scores of Section 5 should be read in that sense. The same caveat applies to (9) and to the standard errors reported for C_{fit}, which come from the Fisher information of a Poisson model whose counts are in truth multinomial and deterministic; all of these measure arithmetic fluctuation, not sampling error in the usual sense.

The power-law exponents assume independent residuals. They are unweighted least-squares fits on nineteen points whose residuals are visibly autocorrelated, so the quoted standard errors on β understate the true uncertainty.

D is a discrepancy, not a test statistic. Only its trend, and the bootstrap uncertainty on that trend, should be read.

9 Conclusion

A complete forward-gap census to 10^{10} takes well under a minute on one core. Using it we find that the fitted decay rate falls monotonically across 20 scales under a reasonable specification, but that its dependence on the fitting window exceeds its dependence on X, and that the one published value traceable to a stated procedure is reproduced from our own census, to the three decimals reported, by a change of specification alone. Neither the plateau reading nor its refutation can be settled this way. Replacing the fitted rate with local, window-free diagnostics obtained by differencing the census, we find that E_2, E_3 and D move toward the Gallagher prediction at 17 of 18 consecutive steps and by 10 to 21 standard errors end to end, with deviations decaying as a fractional power of $\log X$. On the spatial side we derive, from the prime density and the Hardy–Littlewood envelope with no free parameters, the scaling form $R = 1 + \Phi(g/\log X)/\log X$ and hence a crossover at $g \approx \log X$; the location is confirmed to 4% and the form is confirmed by a collapse across 20 scales, while the amplitude is under-predicted by a scale-independent factor of 1.28 ± 0.04 that measures, without any window, how far the actual gap law is from the exponential envelope.

If there is a methodological moral, it is that a diagnostic's robustness to analyst choices matters here as much as the range over which it is computed. Differencing a cumulative census into local, disjoint samples is cheap, general, and removes the window and the scale-mixing problem at once.

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Declaration of interest

The authors report there are no competing interests to declare.

Declaration of generative AI use

The authors used Claude (Anthropic), a generative AI assistant, during the preparation of this work: to help write and check the analysis and verification code, and to edit the text and format the manuscript. It was not used to generate data or results. Every numerical result was produced by the released code and verified as described in Section 3; the authors reviewed and edited all AI-assisted material and take full responsibility for the content of this article.

Data availability statement

The census `fgp_layers.csv`, every derived table and figure, the code that produces them and the Lean source are openly available at https://github.com/farahaymen/Forward-Gap-Layers-of-Consecutive-Primes-Window-Free-Diagnostics-of-the-Gallagher-Limit, released under CC0.

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A The first-order expansion

Substitute $t = e^{L-v}$ in (10), so that $dt = -e^{L-v}dv$ and v runs from 0 to $L - \log 2$. Up to common factors of e^L, the four integrals in (10) become

$$J_1 = \int e^{-2v} \frac{e^{-g/(L-v)}}{(L-v)^2}\, dv, \qquad J_2 = \int e^{-v} \frac{e^{-g/(L-v)}}{(L-v)^2}\, dv,$$
$$J_3 = \int \frac{e^{-2v}}{L-v}\, dv, \qquad J_4 = \int \frac{e^{-v}}{L-v}\, dv,$$

and $R = (J_1/J_2)/(J_3/J_4)$. Write $a = g/L^2$; the regime of interest is $g = xL$ with $x = O(1)$, so $a = x/L$ is $O(1/L)$. Then $e^{-g/(L-v)} = e^{-g/L}e^{-av}(1 + O(v^2/L^2))$ and $(L-v)^{-2} = L^{-2}(1 + 2v/L + O(v^2/L^2))$, and with $\int_0^\infty ve^{-sv}dv = s^{-2}$, the extension of the upper limit to ∞ costing only $O(e^{-L})$,

$$J_1 \propto \tfrac{1}{2}\big(1 - \tfrac{a}{2} + \tfrac{1}{L}\big), \quad J_2 \propto 1 - a + \tfrac{2}{L}, \quad J_3 \propto \tfrac{1}{2L}\big(1 + \tfrac{1}{2L}\big), \quad J_4 \propto \tfrac{1}{L}\big(1 + \tfrac{1}{L}\big),$$

up to $O(L^{-2})$ relative corrections, so $J_1/J_2 = \frac{1}{2}(1 + \frac{a}{2} - \frac{1}{L}) + O(L^{-2})$, $J_3/J_4 = \frac{1}{2}(1 - \frac{1}{2L}) + O(L^{-2})$, and $R = 1 + \frac{a}{2} - \frac{1}{2L} + O(L^{-2})$, which is (11). In x, $L(R-1) = (x-1)/2 + O(1/L)$. The $O(1/L)$ term in the collapsed variable is what Figure 6(c) shows the full model still carrying at $L = 23$ (slope 0.62 rather than 0.5). The numerical model agrees with this expansion: at $\log X = 10^4$ the quantity $\log X\,(R-1)$ matches $(x-1)/2$ to 3×10^{-4}.

B Elementary facts used in the analysis

Four elementary statements are used above. We state them with their proofs; the released file `lean/FGP.lean` contains a Lean 4 formalization of the same statements against a pinned revision of Mathlib [6, 23].

Lemma 1 (Partition)**.** *For $N \geq 2$ the nonempty forward-gap layers $L(g;N)$ are pairwise disjoint and their union is $\{p_2, \ldots, p_N\}$.*

Proof. Each p_n with $2 \leq n \leq N$ lies in $L(d_n;N)$ and in no other layer, since d_n is a single number. □

Lemma 2 (Boundary)**.** *For every $X \geq 2$ and every prime $p \leq X$, the next prime after p is at most $2X$. Hence a sieve of $[2, 2X]$ determines the layer of every prime up to X.*

Proof. By Bertrand's postulate there is a prime r with $p < r \leq 2p \leq 2X$; the next prime after p is at most r. □

Lemma 3 (Telescoping)**.** $\sum_{n=2}^{N} d_n = p_{N+1} - p_2$*, so the mean forward gap over $p_2, \ldots, p_N$ is $(p_{N+1} - p_2)/(N-1)$.*

Proof. The sum telescopes. □

Lemma 4 (Exponential moments)**.** *If Z is exponential with rate $\lambda > 0$, then $\mathbb{E}Z^k = k!/\lambda^k$; in particular $\mathbb{E}Z^2 = 2(\mathbb{E}Z)^2$ and $\mathbb{E}Z^3 = 6(\mathbb{E}Z)^3$, so $E_2 = E_3 = 1$ for an exponential law.*

Proof. $\mathbb{E}Z^k = \int_0^\infty z^k \lambda e^{-\lambda z}\, dz = \lambda^{-k}\Gamma(k+1) = k!/\lambda^k$. □

Lemma 2 is what replaces a numerical argument about the sieve margin by a theorem; Lemma 3 is the identity behind the moment-estimator reading of C_W in Section 4. We do not attempt to formalize the expansion of Appendix A, which would require a substantial development of parameter-dependent integral asymptotics, nor any of the census values, which lie far beyond kernel-checkable computation.

C Register of assumptions

Every assumption the results rest on is listed here with its status: **V** verified by computation (the check is in the released `selftest.py`), **S** stated and relied upon without proof, or **L** a limitation that bounds what is claimed.

The census

- V *The sieve is correct.* Checked against tabulated $\pi(10^k)$ and p_n, the twin-prime counts $\pi_2(10^k)$ (the $g = 2$ layer at 10^{10} must be 27,412,679), and the maximal-gap record; and reproduced independently by a NumPy implementation sharing no code, language or data layout, over the whole range: all 2,166 layer counts at all 20 checkpoints agree exactly to $X = 9{,}999{,}999{,}967$, and the layer sums and sums of squares agree at the smallest checkpoint, where the primes can be enumerated directly.

- V *Layers partition the classified primes.* The counts sum to $N - 1$ at every checkpoint, and the gaps telescope to $p_{N+1} - p_2$, recovering $p_{N+1} = 10{,}000{,}000{,}019$.

- V *No gap is silently dropped.* The accumulator is bounded at $g < 512$; both the census and the independent verifier now abort rather than truncating if a larger gap occurs. Below 10^{10} the maximal gap is 354, but the bound binds above roughly 7×10^{11}.

V *The verifier covers the last checkpoint.* A verifier sieving exactly $[2, X]$ cannot classify its own last prime and so cannot report the checkpoint at $N = \pi(X)$; it sieves past X by a margin, as the census does.

V *No integer overflow.* Layer prime-sums total 2.2×10^{18}, within 64 bits by a factor of eight; sums of squares reach 4.6×10^{28} and are accumulated in 128 bits.

V *Checkpoint/restart does not perturb the result.* Runs interrupted and resumed at segment boundaries produce byte-identical output.

V *Sieving past X suffices to classify every prime up to X.* A margin of 10^3 is used; Bertrand's postulate gives the unconditional bound $2X$ (Lemma 2).

The frequency statistic

V *The arithmetic factor $H(g)$ is computed correctly.* Checked against symbolic factorization for all even $g < 400$; this check detected an incorrect initial implementation (Section 3).

V *The Poisson fit finds the likelihood maximum.* The IRLS slope agrees with a derivative-free optimizer on the same objective to 10^{-7}.

S *Counts are treated as independent Poisson variates.* They are neither independent nor random; the standard errors on C_{fit} are those of that working model and measure arithmetic fluctuation only.

S *The Hardy–Littlewood envelope applies to consecutive gaps.* The singular series governs prime pairs at distance g; its use for the gap to the *next* prime is the standard heuristic step, and the departure it induces is part of what Sections 5 and 6 measure.

V *The comparison with the literature is in the right normalization.* Wolf fits in $u = g\,\pi(X)/X$, so his slope is C_{fit}/C_W and not C_{fit}; (4) records the conversion, and the published value is reproduced from this census to the three decimals reported under the specification his description implies.

L *The fitting window is an analyst choice.* This is the subject of Section 4.4 rather than an assumption we defend.

The local diagnostics

V *Differencing gives the exact distribution on each interval.* The differenced counts are non-negative and sum to the interval's prime count; the local mean gap matches $\log X_{\text{eff}}$ to 5×10^{-4} relative.

V *The discrepancy D is a supremum over the full even grid.* Unoccupied gaps are included, since the survival function is a step function and the supremum can fall at a gap no prime carries. In this range the value is unchanged, because the holes lie in the far tail.

V *The bootstrap standard errors are correct.* They agree with the delta method to within 5% and are stable across seeds.

S *Independence across intervals holds under multinomial resampling.* The intervals are disjoint sets of primes; the primes themselves are deterministic.

V *E_2, E_3 and D have the stated exponential values.* Proved in Lemma 4 and checked by quadrature.

L *The power-law exponents are descriptive.* Unweighted fits on nineteen autocorrelated residuals; the standard errors on β are optimistic.

L *D is convention-dependent.* Its absolute level, and the exponent fitted to it, differ between the left-endpoint and midpoint conventions; only the trend is read.

The spatial statistic

V *The standard error* (9) *is correct.* Checked against a direct bootstrap over the enumerated primes of a layer.

V *The amplitude ratio is measured, not eyeballed.* Census and model slopes are fitted over the same window in the scaling variable at every scale; the ratio is 1.28 ± 0.04 with range $[1.20, 1.39]$ and no trend.

V $C_2 H(g)$ *cancels in R.* It is constant in t and appears in both the numerator and the denominator of the layer mean.

V *The density model is evaluated accurately.* The quadrature is performed in $u = \log t$ so that no overflow occurs, and reproduces the analytic first-order expansion of Appendix A to 3×10^{-4} at $\log X = 10^4$.

S *The model's envelope.* The conditional gap law is taken to be exactly exponential with rate $1/\log t$, and the prime density to be $1/\log t$. Both are approximations; the $\pi(t)/t$ variant is reported alongside, and the resulting amplitude error is measured rather than assumed (Section 6.3).

L *The scales are nested. R* requires the whole range $[2, X]$ and cannot be localized, so successive g^* are strongly correlated and only single-scale statements are made.

L *Layers with fewer than* 30 *primes are excluded from R.* The far tail $g \gtrsim 1.5 \log X$ is therefore not tested.

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Table 1: Independent NumPy census against the C census: layer counts compared at each checkpoint with $X \leq 9{,}999{,}999{,}967$.

checkpoint k	X	layers compared	mismatches
0	2,750,159	62	0
1	4,262,509	65	0
2	6,601,211	70	0
3	10,213,061	74	0
4	15,485,863	77	0
5	24,373,961	81	0
6	37,614,323	85	0
7	57,995,527	93	0
8	89,359,883	96	0
9	160,481,183	103	0
10	211,733,111	108	0
11	325,619,971	112	0
12	500,465,113	119	0
13	768,813,691	127	0
14	1,180,371,793	129	0
15	2,038,074,743	143	0
16	2,778,410,251	146	0
17	4,259,763,509	150	0
18	6,528,037,501	160	0
19	9,999,999,967	166	0

Table 2: The 20-scale census under the primary specification (Poisson GLM, $4 \le g \le 3\log X$). The envelope column is the range of C_{fit} over all 36 specifications at that scale.

N	$X = p_N$	$\log X$	$C_W(X)$	$C_{\mathrm{fit}}(X)$	SE	$\Delta(X)$	envelope	pts
200,000	2,750,159	14.827	1.0783	1.2483	0.00377	0.1700	[1.020, 1.356]	21
300,411	4,262,509	15.265	1.0759	1.2391	0.00312	0.1633	[1.013, 1.342]	21
451,234	6,601,211	15.703	1.0734	1.2311	0.00253	0.1577	[0.999, 1.351]	22
677,778	10,213,061	16.139	1.0711	1.2251	0.00202	0.1541	[1.015, 1.334]	23
1,000,000	15,485,863	16.555	1.0691	1.2160	0.00168	0.1469	[1.007, 1.331]	23
1,529,183	24,373,961	17.009	1.0671	1.2109	0.00134	0.1438	[1.000, 1.316]	24
2,296,917	37,614,323	17.443	1.0651	1.2040	0.00109	0.1389	[0.994, 1.304]	25
3,450,095	57,995,527	17.876	1.0634	1.1967	0.00090	0.1333	[0.988, 1.295]	25
5,182,233	89,359,883	18.308	1.0617	1.1923	0.00072	0.1306	[1.008, 1.282]	26
9,000,000	160,481,183	18.894	1.0596	1.1854	0.00055	0.1258	[1.000, 1.271]	27
11,691,997	211,733,111	19.171	1.0586	1.1817	0.00048	0.1231	[0.996, 1.265]	27
17,562,024	325,619,971	19.601	1.0572	1.1750	0.00039	0.1178	[0.992, 1.259]	28
26,379,128	500,465,113	20.031	1.0558	1.1728	0.00031	0.1170	[1.038, 1.251]	29
39,622,904	768,813,691	20.460	1.0545	1.1674	0.00026	0.1129	[1.034, 1.245]	29
59,515,785	1,180,371,793	20.889	1.0533	1.1646	0.00021	0.1114	[1.030, 1.238]	30
100,000,000	2,038,074,743	21.435	1.0517	1.1584	0.00016	0.1067	[1.025, 1.233]	31
134,277,704	2,778,410,251	21.745	1.0509	1.1550	0.00014	0.1041	[1.022, 1.229]	31
201,692,513	4,259,763,509	22.172	1.0498	1.1528	0.00011	0.1030	[1.007, 1.224]	32
302,953,271	6,528,037,501	22.599	1.0488	1.1485	0.00009	0.0997	[1.003, 1.219]	32
455,052,511	9,999,999,967	23.026	1.0478	1.1460	0.00008	0.0982	[1.000, 1.215]	33

Table 3: Sensitivity of C_{fit} to window and estimator. "mon." records whether C_{fit} decreases (dec.) or increases (inc.) at every one of the 20 scales, and "first" and "last" are its values at the smallest and largest scale. The full grid is in the released file `specification_grid.csv`.

$g \geq 2$						$g \geq 4$					
window	est.	first	last	net	mon.	window	est.	first	last	net	mon.
$1 \log X$	OLS	1.0269	1.0036	-0.0233	–	$1 \log X$	OLS	1.1342	1.0486	-0.0856	–
$1 \log X$	Poisson	1.0197	1.0001	-0.0196	–	$1 \log X$	Poisson	1.1333	1.0476	-0.0857	–
$1 \log X$	WLS	1.0213	0.9999	-0.0214	–	$1 \log X$	WLS	1.1326	1.0462	-0.0864	–
$2 \log X$	OLS	1.1630	1.1277	-0.0352	–	$2 \log X$	OLS	1.1972	1.1443	-0.0529	–
$2 \log X$	Poisson	1.1404	1.1031	-0.0373	–	$2 \log X$	Poisson	1.1877	1.1237	-0.0640	–
$2 \log X$	WLS	1.1402	1.1030	-0.0372	–	$2 \log X$	WLS	1.1858	1.1231	-0.0627	–
$3 \log X$	OLS	1.2728	1.1616	-0.1112	–	$3 \log X$	OLS	1.2937	1.1706	-0.1231	–
$3 \log X$	Poisson	1.2138	1.1314	-0.0824	dec.	$3 \log X$	Poisson	1.2483	1.1460	-0.1023	dec.
$3 \log X$	WLS	1.2122	1.1313	-0.0809	dec.	$3 \log X$	WLS	1.2458	1.1456	-0.1002	dec.
$4 \log X$	OLS	1.3041	1.1888	-0.1152	dec.	$4 \log X$	OLS	1.3174	1.1947	-0.1227	dec.
$4 \log X$	Poisson	1.2356	1.1472	-0.0884	dec.	$4 \log X$	Poisson	1.2664	1.1596	-0.1068	dec.
$4 \log X$	WLS	1.2331	1.1469	-0.0862	dec.	$4 \log X$	WLS	1.2633	1.1590	-0.1043	dec.
$5 \log X$	OLS	1.3449	1.2013	-0.1437	dec.	$5 \log X$	OLS	1.3549	1.2054	-0.1495	dec.
$5 \log X$	Poisson	1.2490	1.1538	-0.0952	dec.	$5 \log X$	Poisson	1.2778	1.1652	-0.1126	dec.
$5 \log X$	WLS	1.2453	1.1533	-0.0921	dec.	$5 \log X$	WLS	1.2741	1.1646	-0.1095	dec.
$6 \log X$	OLS	1.3486	1.2119	-0.1367	–	$6 \log X$	OLS	1.3558	1.2150	-0.1408	–
$6 \log X$	Poisson	1.2517	1.1572	-0.0945	dec.	$6 \log X$	Poisson	1.2797	1.1682	-0.1115	dec.
$6 \log X$	WLS	1.2479	1.1565	-0.0915	dec.	$6 \log X$	WLS	1.2759	1.1674	-0.1085	dec.

Table 4: Window-free local diagnostics on the nineteen differenced intervals. E_2, E_3 equal 1 and D equals 0 for an exponential law; D uses the left-endpoint convention. The column $\log X_{\mathrm{eff}}/m_1$ is a local test of the Prime Number Theorem.

k	X_{k-1}	X_k	gaps	m_1	$\log X_{\mathrm{eff}}/m_1$	E_2	E_3	D
1	2,750,159	4,262,509	100,411	15.0616	1.00004	0.84606	0.68323	0.05643
2	4,262,509	6,601,211	150,823	15.5063	0.99959	0.84574	0.68190	0.05767
3	6,601,211	10,213,061	226,544	15.9434	0.99959	0.85066	0.69260	0.05581
4	10,213,061	15,485,863	322,222	16.3638	0.99987	0.85195	0.69415	0.05521
5	15,485,863	24,373,961	529,183	16.7959	1.00020	0.85602	0.70253	0.05483
6	24,373,961	37,614,323	767,734	17.2460	0.99974	0.85804	0.70583	0.05417
7	37,614,323	57,995,527	1,153,178	17.6739	1.00006	0.86189	0.71499	0.05339
8	57,995,527	89,359,883	1,732,138	18.1073	1.00001	0.86536	0.72097	0.05227
9	89,359,883	160,481,183	3,817,767	18.6290	1.00002	0.86792	0.72633	0.05148
10	160,481,183	211,733,111	2,691,997	19.0386	1.00000	0.87070	0.73230	0.05084
11	211,733,111	325,619,971	5,870,027	19.4014	1.00000	0.87232	0.73538	0.05020
12	325,619,971	500,465,113	8,817,104	19.8302	1.00006	0.87457	0.73991	0.04952
13	500,465,113	768,813,691	13,243,776	20.2622	0.99994	0.87689	0.74453	0.04882
14	768,813,691	1,180,371,793	19,892,881	20.6887	1.00006	0.87864	0.74798	0.04822
15	1,180,371,793	2,038,074,743	40,484,215	21.1861	1.00004	0.88105	0.75266	0.04743
16	2,038,074,743	2,778,410,251	34,277,704	21.5982	1.00000	0.88292	0.75633	0.04676
17	2,778,410,251	4,259,763,509	67,414,809	21.9737	1.00001	0.88478	0.76026	0.04621
18	4,259,763,509	6,528,037,501	101,260,758	22.4003	1.00003	0.88638	0.76339	0.04566
19	6,528,037,501	9,999,999,967	152,099,240	22.8270	1.00003	0.88815	0.76691	0.04505

Table 5: Bootstrap standard errors at six representative intervals; all nineteen are in `local_diagnostics.csv`.

k	$\log X_{\mathrm{eff}}$	E_2 (SE)	E_3 (SE)	D_{left} (SE)	D_{mid} (SE)
1	15.062	0.84606 (0.00201)	0.68323 (0.00507)	0.05643 (0.00109)	0.10569 (0.00110)
5	16.799	0.85602 (0.00093)	0.70253 (0.00241)	0.05483 (0.00045)	0.10038 (0.00045)
9	18.629	0.86792 (0.00037)	0.72633 (0.00097)	0.05148 (0.00017)	0.09365 (0.00017)
13	20.261	0.87689 (0.00020)	0.74453 (0.00055)	0.04882 (0.00009)	0.08835 (0.00009)
16	21.598	0.88292 (0.00013)	0.75633 (0.00036)	0.04676 (0.00005)	0.08436 (0.00005)
19	22.828	0.88815 (0.00006)	0.76691 (0.00017)	0.04505 (0.00002)	0.08102 (0.00002)

Table 6: Consecutive-step and end-to-end tests. A step counts as "toward the limit" when the statistic moves toward its exponential value; z is the change divided by the root-sum-square of the two bootstrap standard errors. β is the fitted exponent in $(\log X)^{-\beta}$.

statistic	steps toward limit	median z	min z	end-to-end change	end-to-end z	exponent β
E_2	17/18	4.5	-0.12	+0.0421 $\pm$ 0.0020	21	0.810 $\pm$ 0.011
E_3	17/18	3.7	-0.20	+0.0837 $\pm$ 0.0051	17	0.782 $\pm$ 0.012
D_{left}	17/18	2.8	-0.88	-0.0114 $\pm$ 0.0011	10	0.592 $\pm$ 0.017
D_{mid}	17/18	5.5	-0.17	-0.0247 $\pm$ 0.0011	22	0.666 $\pm$ 0.010
W^2	18/18	6.0	0.47	-0.001199 $\pm$ 0.000048	25	–

Table 7: The computed crossover under the six unweighted locators and the primary weighted locator. The bold column is the unweighted fit over the primary window $x \in [0.6, 1.4]$; the last two columns are the weighted fit over the same window.

$X = p_N$	$\log X$	window in x			band in R			weighted fit	
		$[0.7, 1.3]$	$[0.6, 1.4]$	$[0.5, 1.5]$	0.02	0.03	0.05	g^* (WLS)	$\log X - g^*$
2,750,159	14.827	13.79	**14.26**	14.57	13.54	14.57	14.18	14.13 ± 0.45	0.69
4,262,509	15.265	14.29	**14.54**	15.03	14.29	14.82	14.51	14.55 ± 0.39	0.71
6,601,211	15.703	14.52	**14.98**	15.48	14.49	15.48	15.15	14.94 ± 0.38	0.76
10,213,061	16.139	15.05	**15.44**	15.37	15.52	15.48	15.25	15.49 ± 0.43	0.65
15,485,863	16.555	15.49	**16.00**	15.96	15.72	15.84	15.63	15.89 ± 0.43	0.66
24,373,961	17.009	16.11	**16.11**	16.28	16.11	16.18	16.07	16.11 ± 0.58	0.89
37,614,323	17.443	16.65	**16.44**	16.56	16.48	16.62	16.49	16.38 ± 0.34	1.06
57,995,527	17.876	16.97	**16.88**	17.10	16.91	17.03	16.79	16.87 ± 0.30	1.01
89,359,883	18.308	17.84	**17.66**	17.69	17.66	17.62	17.31	17.66 ± 0.39	0.65
160,481,183	18.894	18.20	**18.12**	18.31	18.12	18.16	17.84	18.12 ± 0.21	0.77
211,733,111	19.171	18.51	**18.38**	18.56	18.44	18.42	18.09	18.36 ± 0.25	0.81
325,619,971	19.601	19.01	**18.94**	19.11	19.01	18.93	18.54	18.95 ± 0.29	0.65
500,465,113	20.031	19.28	**19.53**	19.43	19.54	19.23	18.98	19.55 ± 0.38	0.48
768,813,691	20.460	19.81	**19.96**	19.84	19.93	19.52	19.38	19.97 ± 0.33	0.49
1,180,371,793	20.889	20.15	**20.29**	20.19	20.23	19.91	19.77	20.30 ± 0.32	0.58
2,038,074,743	21.435	20.74	**20.82**	20.68	20.82	20.51	20.32	20.81 ± 0.28	0.62
2,778,410,251	21.745	21.20	**21.16**	21.02	21.18	20.78	20.65	21.14 ± 0.28	0.61
4,259,763,509	22.172	21.76	**21.69**	21.48	21.54	21.23	21.09	21.65 ± 0.27	0.52
6,528,037,501	22.599	22.17	**22.04**	21.86	21.88	21.68	21.52	22.00 ± 0.27	0.60
9,999,999,967	23.026	22.56	**22.34**	22.31	22.28	22.09	21.94	22.37 ± 0.27	0.66

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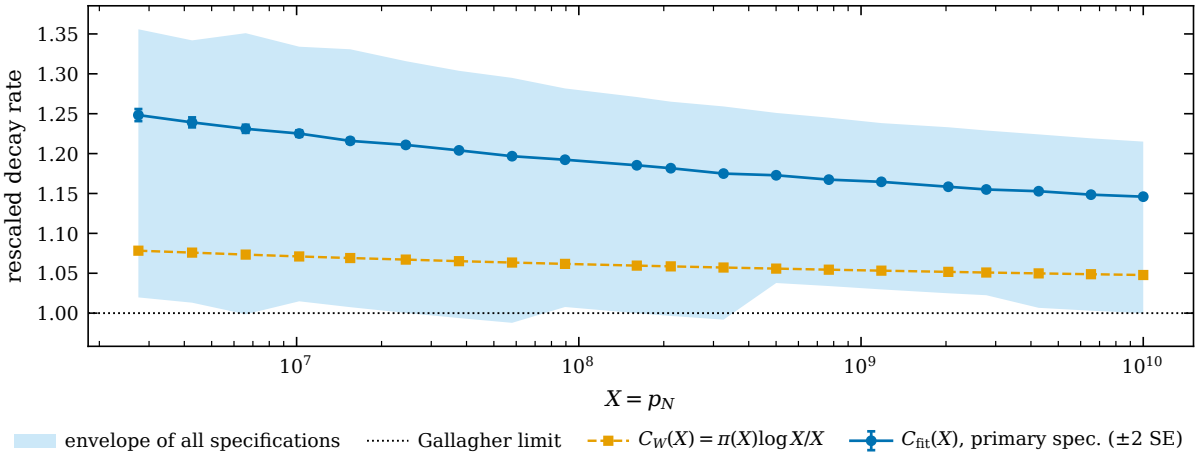


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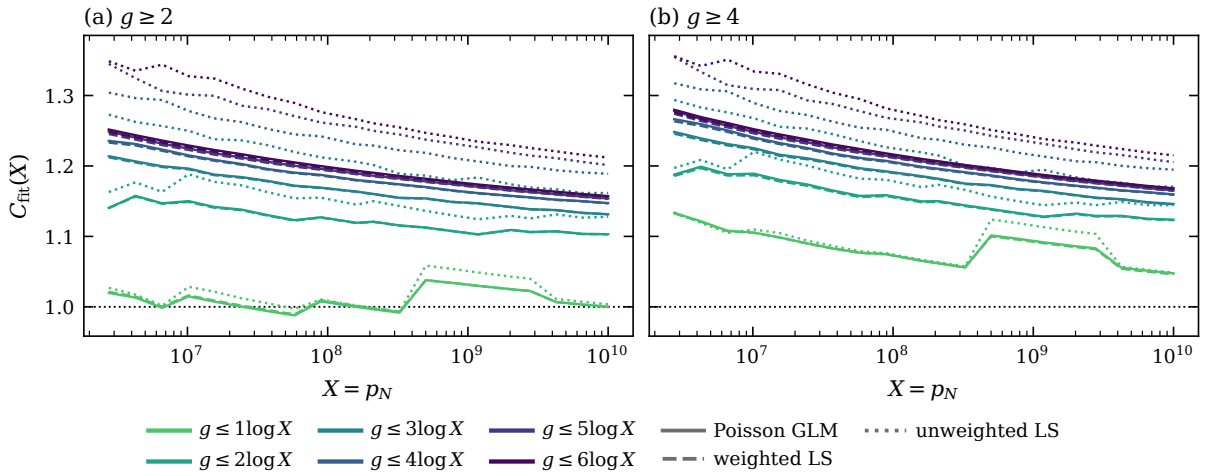


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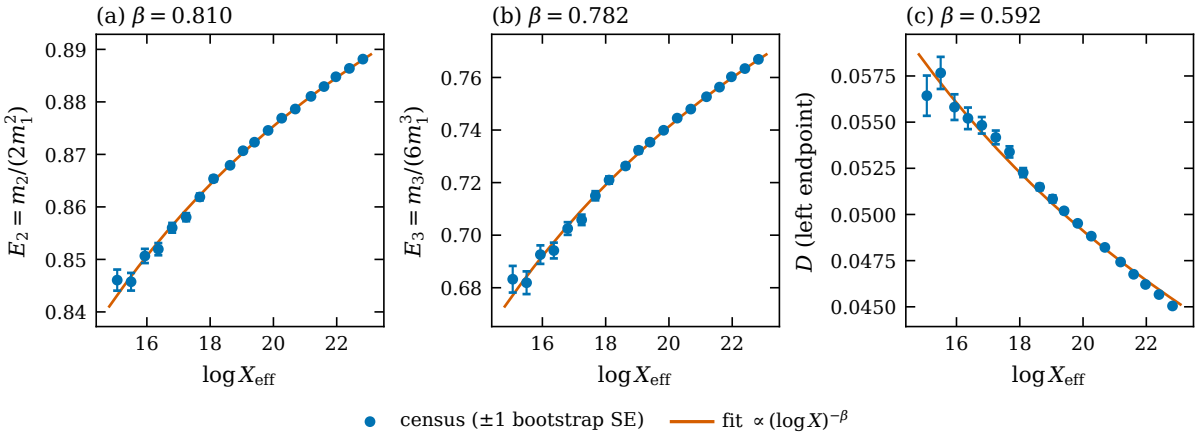


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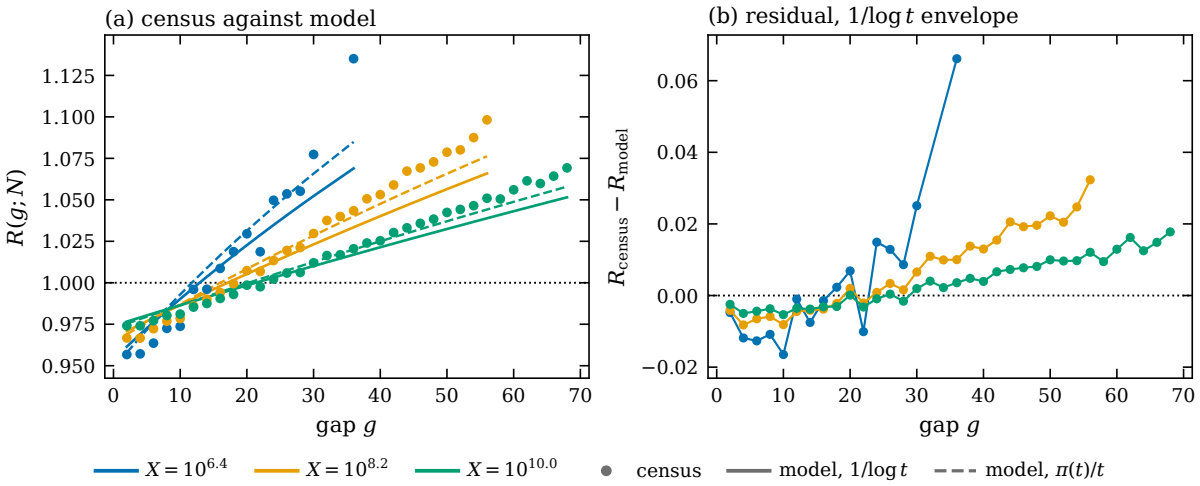


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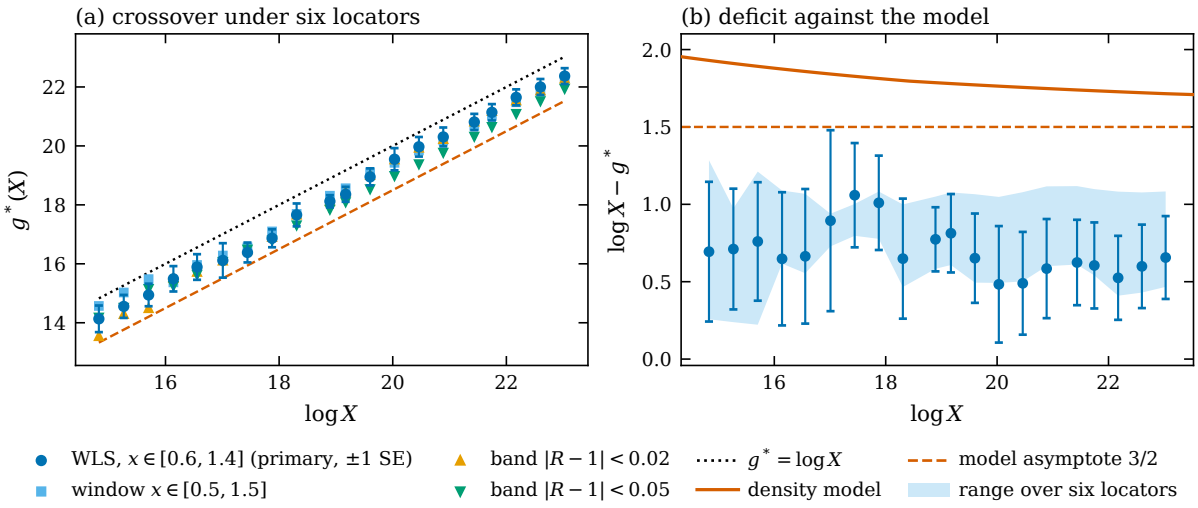


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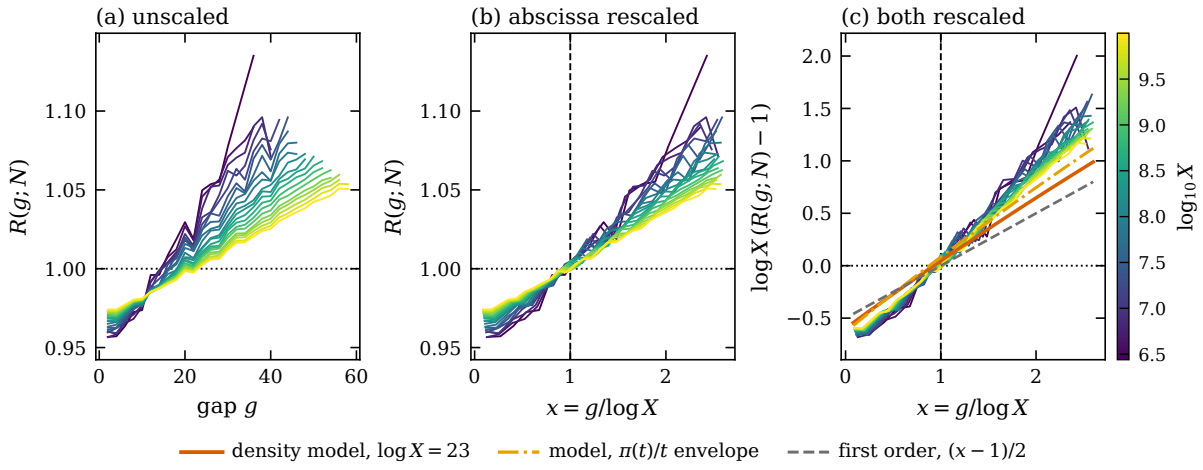


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